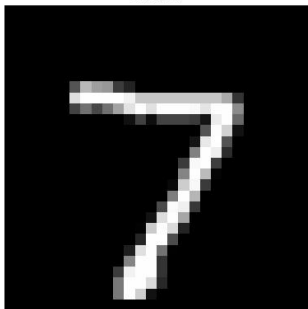




|                  |                                 |
|------------------|---------------------------------|
| NAMA :           | MUHAMMAD NABIRULLAH BIN ZUKAIRI |
| KELAS :          | 1 DKA SEM 2                     |
| TARIKH           | 27/8/2026                       |
| TAJUK :          | LAPORAN AMALI 1                 |
| NAMA PENSYARAH : | SARASWATHY A/P CHANDIRAN        |

| LANGKAH                                                                                                                                                                                                                                                                           | JAWAPAN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. LANGKAH 1 - Sediakan folder projek dan Git</b>                                                                                                                                                                                                                              | <b>Langkah 1 — Sediakan Folder Projek &amp; Git</b> <ul style="list-style-type: none"> <li>Cipta folder dengan nama : DKA3223_AMALI1_NABIL</li> <li>Cipta repository di GitHub</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>2. LANGKAH 2 - Import library</b> <ul style="list-style-type: none"> <li>Import library yang diperlukan untuk tensor, neural network, dataset, DataLoader dan paparan imej.</li> </ul> <div> <b>Contoh output semakan:</b><br/> PyTorch version: 2.x.x<br/> Device: cpu </div> | <pre> import torch import torch.nn as nn import torch.optim as optim from torch.utils.data import DataLoader from torchvision import datasets, transforms import matplotlib.pyplot as plt  # Semak versi PyTorch dan peranti yang digunakan (CPU/GPU) device = torch.device("cuda" if torch.cuda.is_available() else "cpu") print("PyTorch version:", torch.__version__) print("Device:", device) </pre> <pre> ... PyTorch version: 2.11.0+cpu Device: cpu </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>3. LANGKAH 3 - Muatkan dan semak dataset MNIST</b> <div> <b>Contoh output semakan:</b><br/> Training data: 60000<br/> Testing data: 10000<br/> Image shape: torch.Size([1, 28, 28])<br/> Actual label: 7<br/> [Paparan satu imej digit MNIST] </div>                           | <pre> # 3.3.1 - Transform: tukar imej MNIST kepada tensor transform = transforms.ToTensor()  # 3.3.2 - Muatkan training &amp; testing dataset train_dataset = datasets.MNIST(root=".", train=True, download=True, transform=transform) test_dataset = datasets.MNIST(root=".", train=False, download=True, transform=transform)  # 3.3.3 - Bina DataLoader dengan batch_size = 64 train_loader = DataLoader(train_dataset, batch_size=64, shuffle=True) test_loader = DataLoader(test_dataset, batch_size=64, shuffle=False)  # 3.3.4 - Papar bilangan data print("Training data:", len(train_dataset)) print("Testing data:", len(test_dataset))  # 3.3.5 - Ambil satu imej daripada testing dataset &amp; paparkan image, label = test_loader.dataset[0] print("Image shape:", image.shape) print("Actual label:", label)  plt.imshow(image.squeeze(), cmap="gray") plt.title(f"Label: {label}") plt.axis("off") plt.show() </pre> <pre> ... Training data: 60000 Testing data: 10000 Image shape: torch.Size([1, 28, 28]) Actual label: 7 </pre> <p style="text-align: center;">Label: 7</p>  |
| <b>4. LANGKAH 4 - Bina model ANN</b> <div> <b>Contoh bentuk output:</b><br/> Flatten(start_dim=1)<br/> Linear(in_features=784, out_features=...)<br/> ReLU()<br/> Linear(in_features=..., out_features=10) </div>                                                                 | <pre> class ANN(nn.Module):     def __init__(self):         super(ANN, self).__init__()         self.flatten = nn.Flatten()         self.fc1 = nn.Linear(28*28, 128)  # input 784 -&gt; hidden layer 128         self.relu = nn.ReLU()         self.fc2 = nn.Linear(128, 10)     # hidden layer -&gt; output 10 kelas      def forward(self, x):         x = self.flatten(x)         x = self.fc1(x)         x = self.relu(x)         x = self.fc2(x)         return x  ann_model = ANN().to(device) print(ann_model)  # 3.4.4 - Loss function &amp; optimizer criterion_ann = nn.CrossEntropyLoss() optimizer_ann = optim.Adam(ann_model.parameters(), lr=0.001) </pre> <pre> ANN(   (flatten): Flatten(start_dim=1, end_dim=-1)   (fc1): Linear(in_features=784, out_features=128, bias=True)   (relu): ReLU()   (fc2): Linear(in_features=128, out_features=10, bias=True) ) </pre>                                                                                                                                                                                                                                                                                               |

## 5. LANGKAH 5 - Latih dan uji ANN

Contoh output semakan (nilai sebenar boleh berbeza):

Epoch 1/2 - Loss: 0.xxxx  
Epoch 2/2 - Loss: 0.xxxx  
ANN Testing Accuracy: xx.xx%

```
# 3.5.1 - Latih ANN selama 2 epoch
num_epochs = 2

for epoch in range(num_epochs):
    ann_model.train()
    running_loss = 0.0

    for images, labels in train_loader:
        images, labels = images.to(device), labels.to(device)

        optimizer_ann.zero_grad()
        outputs = ann_model(images)
        loss = criterion_ann(outputs, labels)
        loss.backward()
        optimizer_ann.step()

        running_loss += loss.item()

# 3.5.2 - Papar purata loss selepas setiap epoch
avg_loss = running_loss / len(train_loader)
print(f"Epoch {epoch+1}/{num_epochs} - Loss: {avg_loss:.4f}")

Epoch 1/2 - Loss: 0.3461
Epoch 2/2 - Loss: 0.1581

# 3.5.3 - Tukar ke evaluation mode & kira accuracy
ann_model.eval()
correct = 0
total = 0

with torch.no_grad():
    for images, labels in test_loader:
        images, labels = images.to(device), labels.to(device)
        outputs = ann_model(images)
        _, predicted = torch.max(outputs, 1)

        total += labels.size(0)
        correct += (predicted == labels).sum().item()

# 3.5.4 - Simpan accuracy untuk perbandingan nanti
ann_accuracy = 100 * correct / total
print(f"ANN Testing Accuracy: {ann_accuracy:.2f}%")

... ANN Testing Accuracy: 96.02%
```

## 6. LANGKAH 6 - Bina model CNN

Contoh bentuk output:

Input batch: [64, 1, 28, 28]  
Selepas pooling: [64, ..., 14, 14]  
Output model: [64, 10]

```
class CNN(nn.Module):
    def __init__(self):
        super(CNN, self).__init__()
        # Input 1 channel (grayscale MNIST), output 16 feature maps
        self.conv1 = nn.Conv2d(1, 16, kernel_size=3, padding=1)
        self.relu = nn.ReLU()
        self.pool = nn.MaxPool2d(2, 2)

        # Fully connected: selepas pooling, saiz imej jadi 14x14 dengan 16 channel
        self.flatten = nn.Flatten()
        self.fc1 = nn.Linear(16 * 14 * 14, 128)
        self.fc2 = nn.Linear(128, 10)

    def forward(self, x):
        x = self.conv1(x)          # [batch, 16, 28, 28]
        x = self.relu(x)
        x = self.pool(x)           # [batch, 16, 14, 14]
        x = self.flatten(x)        # [batch, 16*14*14]
        x = self.fc1(x)
        x = self.relu(x)
        x = self.fc2(x)
        return x

cnn_model = CNN().to(device)
print(cnn_model)

# 3.6.4 - Loss function & optimizer
criterion_cnn = nn.CrossEntropyLoss()
optimizer_cnn = optim.Adam(cnn_model.parameters(), lr=0.001)

... CNN(
  (conv1): Conv2d(1, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (relu): ReLU()
  (pool): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
  (flatten): Flatten(start_dim=1, end_dim=-1)
  (fc1): Linear(in_features=3136, out_features=128, bias=True)
  (fc2): Linear(in_features=128, out_features=10, bias=True)
)
```

## 7. LANGKAH 7 - Latih dan uji CNN

Contoh output semakan (nilai sebenar boleh berbeza):

Epoch 1/2 - Loss: 0.xxxx  
Epoch 2/2 - Loss: 0.xxxx  
CNN Testing Accuracy: xx.xx%

```
# 3.7.1 - Latih CNN selama 2 epoch
num_epochs = 2

for epoch in range(num_epochs):
    cnn_model.train()
    running_loss = 0.0

    for images, labels in train_loader:
        images, labels = images.to(device), labels.to(device)

        optimizer_cnn.zero_grad()
        outputs = cnn_model(images)
        loss = criterion_cnn(outputs, labels)
        loss.backward()
        optimizer_cnn.step()

        running_loss += loss.item()

# 3.7.2 - Papar purata loss selepas setiap epoch
avg_loss = running_loss / len(train_loader)
print(f"Epoch {epoch+1}/{num_epochs} - Loss: {avg_loss:.4f}")

Epoch 1/2 - Loss: 0.2530
Epoch 2/2 - Loss: 0.0786
```

```
# 3.7.3 - Tukar ke evaluation mode & kira accuracy
cnn_model.eval()
correct = 0
total = 0

with torch.no_grad():
    for images, labels in test_loader:
        images, labels = images.to(device), labels.to(device)
        outputs = cnn_model(images)
        _, predicted = torch.max(outputs, 1)

        total += labels.size(0)
        correct += (predicted == labels).sum().item()

# 3.7.4 - Simpan accuracy untuk perbandingan
cnn_accuracy = 100 * correct / total
print(f"CNN Testing Accuracy: {cnn_accuracy:.2f}%")

... CNN Testing Accuracy: 97.93%
```

## 8. LANGKAH 8 - Uji pengecaman lima imej digit

Contoh output bagi satu imej:

Actual: 7  
ANN Prediction: 7 [BETUL]  
CNN Prediction: 7 [BETUL]  
[Paparan imej digit 7]

```
# 3.8.1 - Ambil 5 imej daripada testing dataset
n_show = 5
fig = plt.figure(figsize=(12, 4))

ann_model.eval()
cnn_model.eval()

with torch.no_grad():
    for i in range(n_show):
        image, actual_label = test_dataset[i]
        image_batch = image.unsqueeze(0).to(device) # tambah dimensi batch: [1, 1, 28, 28]

        # 3.8.2 - Ramalan ANN
        output_ann = ann_model(image_batch)
        _, pred_ann = torch.max(output_ann, 1)
        pred_ann = pred_ann.item()

        # Ramalan CNN
        output_cnn = cnn_model(image_batch)
        _, pred_cnn = torch.max(output_cnn, 1)
        pred_cnn = pred_cnn.item()

        # 3.8.4 - Tanda betul/salah
        status_ann = "[BETUL]" if pred_ann == actual_label else "[SALAH]"
        status_cnn = "[BETUL]" if pred_cnn == actual_label else "[SALAH]"

        # 3.8.3 - Papar imej + label
        ax = fig.add_subplot(1, n_show, i + 1)
        ax.imshow(image.squeeze(), cmap="gray")
        ax.set_title(f"Actual: {actual_label}\nANN: {pred_ann} {status_ann}\nCNN: {pred_cnn} {status_cnn}",
                    fontsize=8)
        ax.axis("off")

    print(f"Actual: {actual_label}")
    print(f"ANN Prediction: {pred_ann} {status_ann}")
    print(f"CNN Prediction: {pred_cnn} {status_cnn}")
    print("----")

plt.tight_layout()
plt.show()

plt.tight_layout()
plt.show()

... Actual: 7
ANN Prediction: 7 [BETUL]
CNN Prediction: 7 [BETUL]
...
Actual: 2
ANN Prediction: 2 [BETUL]
CNN Prediction: 2 [BETUL]
...
Actual: 1
ANN Prediction: 1 [BETUL]
CNN Prediction: 1 [BETUL]
...
Actual: 0
ANN Prediction: 0 [BETUL]
CNN Prediction: 0 [BETUL]
...
Actual: 4
ANN Prediction: 4 [BETUL]
CNN Prediction: 4 [BETUL]
...

Actual 7 ANN: 7 [BETUL] CNN: 7 [BETUL]
Actual 2 ANN: 2 [BETUL] CNN: 2 [BETUL]
Actual 1 ANN: 1 [BETUL] CNN: 1 [BETUL]
Actual 0 ANN: 0 [BETUL] CNN: 0 [BETUL]
Actual 4 ANN: 4 [BETUL] CNN: 4 [BETUL]
```

## 9. LANGKAH 9 - Bandingkan keputusan ANN dan CNN

| MODEL | EPOCH | TESTING ACCURACY | RAMALAN BETUL / 5 |
|-------|-------|------------------|-------------------|
| ANN   | 2     |                  |                   |
| CNN   | 2     |                  |                   |

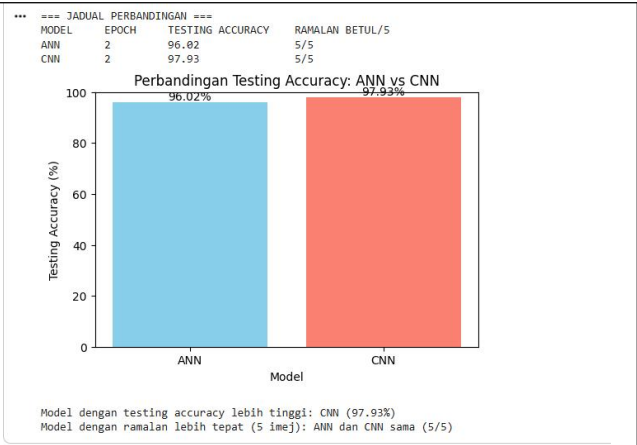
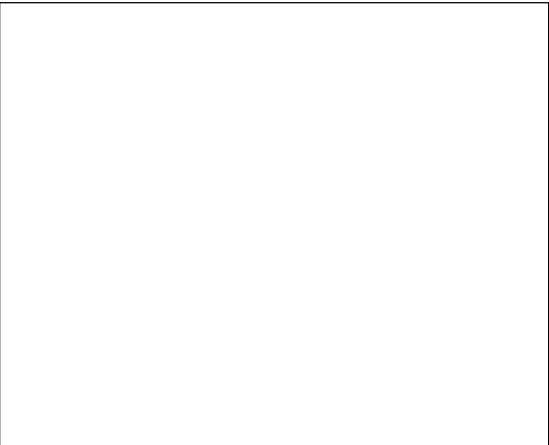
```
# 3.9.1 - Jadual perbandingan accuracy
print("== JADUAL PERBANDINGAN ==")
print(f"Model: '{MODEL}' Epoch: {EPOCH} Testing Accuracy: {TESTING_ACCURACY:.2f} Ramalan Betul: {RAMALAN_BETUL}/5")
print(f"ANN: '{ANN}' Epoch: {EPOCH} Testing Accuracy: {ANN_ACCURACY:.2f} Ramalan Betul: {ANN_BETUL}/5")
print(f"CNN: '{CNN}' Epoch: {EPOCH} Testing Accuracy: {CNN_ACCURACY:.2f} Ramalan Betul: {CNN_BETUL}/5")

# 3.9.2 - Graf bar perbandingan
models = ['ANN', 'CNN']
accuracies = [ann_accuracy, cnn_accuracy]

plt.figure(figsize=(6, 4))
plt.bar(models, accuracies, color=['skyblue', 'salmon'])
plt.title("Perbandingan Testing Accuracy: ANN vs CNN")
plt.xlabel("Model")
plt.ylabel("Testing Accuracy (%)")
plt.ylim(0, 100)
for i, acc in enumerate(accuracies):
    plt.text(i, acc + 1, f"{acc:.2f}%", ha="center")
plt.show()

# 3.9.3 - Model dengan testing accuracy lebih tinggi
if ann_accuracy > cnn_accuracy:
    print(f"Model dengan testing accuracy lebih tinggi: ANN ({ann_accuracy:.2f}%)")
elif cnn_accuracy > ann_accuracy:
    print(f"Model dengan testing accuracy lebih tinggi: CNN ({cnn_accuracy:.2f}%)")
else:
    print(f"Kedua-dua model mempunyai testing accuracy yang sama ({ann_accuracy:.2f}%)")

# 3.9.4 - Model dengan ramalan 5 imej lebih banyak betul
if betul_ann > betul_cnn:
    print(f"Model dengan ramalan lebih tepat (5 imej): ANN ({betul_ann}/5)")
elif betul_cnn > betul_ann:
    print(f"Model dengan ramalan lebih tepat (5 imej): CNN ({betul_cnn}/5)")
else:
    print(f"Model dengan ramalan lebih tepat (5 imej): ANN dan CNN sama ({betul_ann}/5)")
```



**LANGKAH 10: REKOD PENGGUNAAN AI CODE ASSISTANT**

Pautan link Github:

[https://github.com/Brooo-Sapenabil/DKA3223\\_AMALI1\\_NABIRULLAH](https://github.com/Brooo-Sapenabil/DKA3223_AMALI1_NABIRULLAH)

-Bahan bahan disimpan di dalam repo, Github

“DKA3223\_AMALI1\_NABIRULLAH”

